Assignment 1

Pseudocode :

```
Start
Display "Welcome to [Your NamE]'s (Student Number) Number list test!"
Repeat
Ask user for difficulty : "[E]asy, [M]edium, [H]ard"
Set input = user input lowercase.
If input is not "e", "m", or "h" Then
    display "Invalid Choice ! Enter E, M or H."
Until input is valid.

If difficulty = "e" then
Set questions = 3, quantity = 3, minimum = 1, maximum = 10
Else if difficulty = "m" then
Set questions = 5, quantity = 4, minimum = 5, maximum = 15
else
Set questions = 7, quantity = 6, minimum = 10, maximum = 50

Display "Your test will have [questions] questions using list of
[quantity] numbers."
Display "Press Enter to begin!"
Wait for user to press Enter.

Set score = 0
Set start_time = current time.

For each current_q from 1 to questions.
If current_q = questions then.
    Set list = call random_list (quantity * 2, minimum, maximum)
~~Display~~Display "[!!! Challenge Question [!!!]"
else
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    Set list = Call random_list (quantity, minimum, maximum)

Display "Question [current_q] of [questions]"
Display list

Set question_type = random number between 1 and 4

If question_type = 1 then
    Set question_text = "What is the sum of the numbers in the list?"
    Set answer = Sum of all numbers in list
else if question_type = 2 then
    Set question_text = "What is the difference between the minimum
                         and maximum numbers?"
    Set answer = Max(list) - Min(list)
else if question_type = 3 then
    Set average = sum(list) / length(list)
    Set average_floor = FLOOR(average)
    Set question_text = "How many numbers are higher than the
                         average ([average_floor])?"
    Set answer = count of items in list > average_floor
else
    Generate POS1 and POS2 randomly between 1 and length(list)
while POS1 = POS2
    Generate pos 2 randomly
    Set question_text = "What is the product of the numbers in
                         Positions [POS1] and [POS2]?"
    Set answer = list [POS1-1] * list [POS2-1]
    Set result = Call ask_question(question_text, answer)

If result = True then
    Set Score = Score + 1
else
    Set Score = Score - 1
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If Score <0 then
    Set Score =0
End For.

Set total_time = current time - Start_time
Set percentage = Round ((Score/questions) * 100)
Set avg_speed = Round (total_time /questions, 2)

Display "Test complete! You took [Round (total_time,2)] seconds."
Display "You scored [Percentage] % ([Score]/[questions] correct)."
Display "Average of [avg_speed] seconds per question."

If Score = questions then
    Display "perfect score! welldone!"

End.

Function random_list (quantity, min, max)
Create empty list
Repeat quantity times
Add random number between min and max to list
Return list.
End Function

Function ask_question (question_text, answer)
Display question_text
Repeat
Ask user for input
If input is not a valid integer then
Display "Please enter a whole number."
Until input is a valid integer.
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If input = answer then
   Display "Correct!"
   Return True
Else
   Display "Incorrect! Correct answer was [answer]."
   Return False
End Function
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